

National University of Computer & Emerging Sciences Islamabad

FAST School of Computing

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Islamabad Campus

MT1004 – Linear Algebra Homework # 4

Question 1 — Subspaces

A machine-learning system represents each student using a feature vector

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \in \mathbb{R}^4,$$

where the features correspond to four normalized measurements.

Consider the set

$$W = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \in \mathbb{R}^4 : x_1 + x_2 - x_3 = 0, \quad 2x_1 + 2x_2 - 2x_3 = 0 \right\}.$$

1. Determine whether W is a subspace of $\mathbb{R}^4$.
2. Find a basis for W .
3. Find $\dim(W)$.
4. Explain why the two given constraints do not necessarily reduce the dimension by 2.

Question 2 — Matrix Subspaces

Suppose an image-processing algorithm uses 2×2 matrices as filters. Consider

$$V = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} : a + d = 0, \quad b = c \right\}.$$

1. Determine whether V is a subspace of $M_{2 \times 2}$.
2. Find a basis for V .
3. Determine $\dim(V)$.
4. Is the following set a basis for V ?

$$S = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix} \right\}.$$

Justify your answer carefully.

Question 3 — Polynomial Subspaces

A software engineer models execution-time correction functions using polynomials in P_4 . Consider

$$W = \{p(x) \in P_4 : p(1) = 0, \quad p(-1) = 0\}.$$

1. Show whether W is a subspace of P_4 .
2. Find a basis for W .
3. Determine $\dim(W)$.

4. Without expanding every possible polynomial unnecessarily, express the general form of every polynomial in W .
5. Determine whether

$$p(x) = x^4 - x^2$$

belongs to W .

Question 4 — Basis and Linear Independence

A debugging system encodes four types of software events as vectors:

$$v_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 3 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 2 \\ 4 \\ 3 \\ 7 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 1 \\ 2 \\ 2 \\ 4 \end{bmatrix}, \quad v_4 = \begin{bmatrix} 3 \\ 6 \\ 4 \\ 10 \end{bmatrix}.$$

Let

$$W = \text{span}\{v_1, v_2, v_3, v_4\}.$$

1. Find a basis for W by removing redundant vectors.
2. Determine $\dim(W)$.
3. Determine whether v_4 contributes a new direction to the span.
4. Explain the significance of linear dependence here in terms of storing redundant feature information.

Question 5 — Extending to a Basis

In digital communications (like Hamming codes), valid data sequences are often built using linearly independent basis vectors in $\mathbb{R}^4$. Consider the following two linearly independent "message" vectors in $\mathbb{R}^4$:

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}$$

Extend the set $\{v_1, v_2\}$ to form a basis for $\mathbb{R}^4$ by finding two additional vectors, v_3 and v_4 , that are not in the span of the previous vectors.

Question 6 — The Four Fundamental Subspaces

Consider the following 4×5 matrix A :

$$A = \begin{bmatrix} 1 & -1 & 2 & 0 & 3 \\ -2 & 2 & -4 & 1 & -5 \\ 0 & 0 & 0 & 2 & 2 \\ 3 & -3 & 6 & -1 & 8 \end{bmatrix}$$

- (i) Reduce A to row echelon form and determine its rank.
- (ii) Find a basis for the column space $\text{Col}(A)$.
- (iii) Find a basis for the row space $\text{Row}(A)$.
- (iv) Find a basis for the null space $\text{Nul}(A) = \{x \in \mathbb{R}^5 : Ax = 0\}$.
- (v) Find a basis for the left null space $\text{Nul}(A^\top) = \{y \in \mathbb{R}^4 : A^\top y = 0\}$.
- (vi) Verify that the dimensions satisfy the fundamental theorem of linear algebra relations:

$$\dim(\text{Col}(A)) + \dim(\text{Nul}(A)) = n$$

$$\dim(\text{Row}(A)) + \dim(\text{Nul}(A^\top)) = m$$

- (vii) The `sympy` library in Python can compute exact symbolic linear algebra. Write a Python script using `sympy.Matrix` to define matrix A and automatically compute its Reduced Row Echelon Form (RREF) and a basis for its null space.